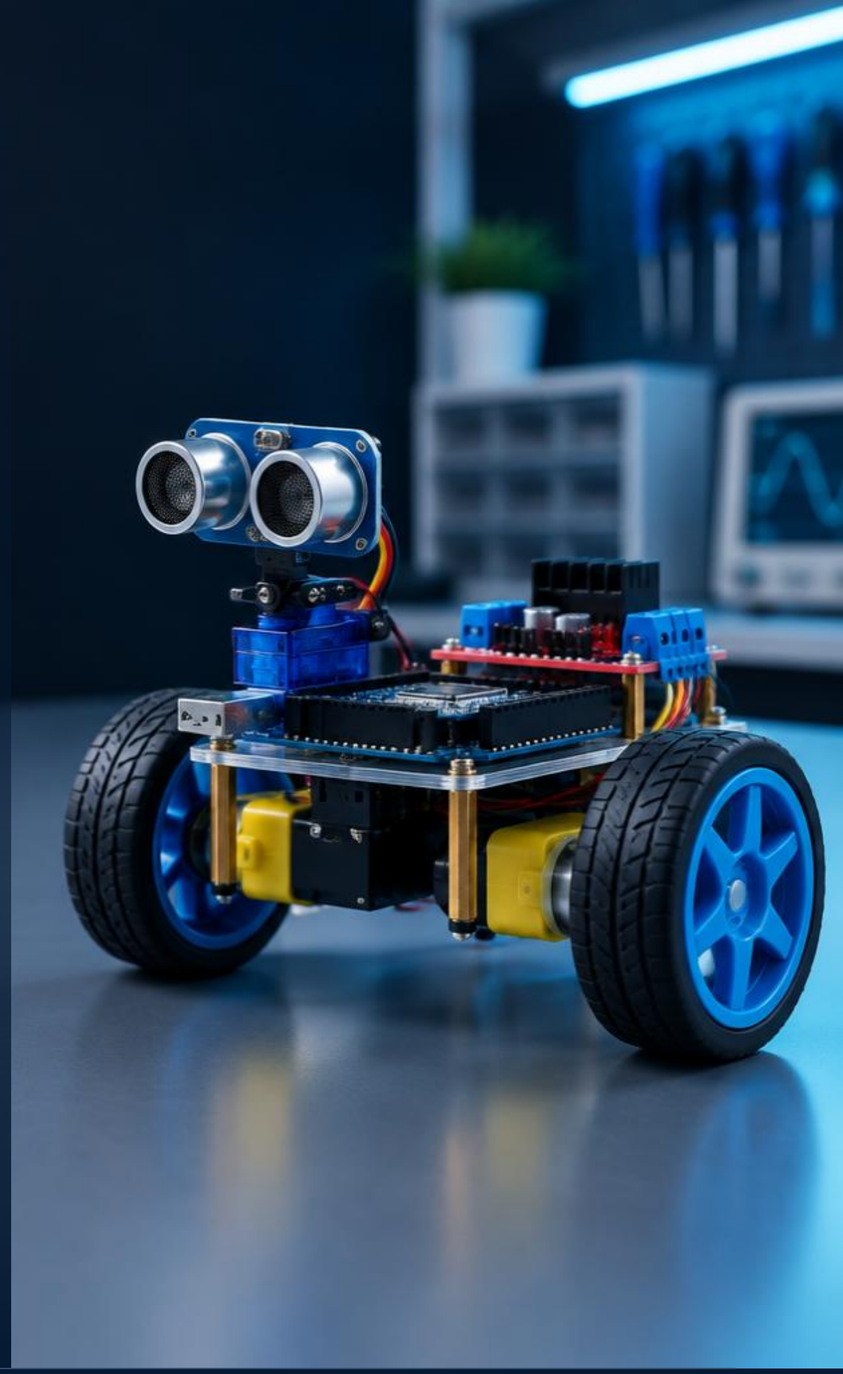
Implementation of an Obstacle-Avoiding Robot

Arduino • Ultrasonic Sensor • L298N Motor Driver



CSE2102 | Microprocessor and Microcontrollers Lab

Department of Computer Science and Engineering
Begum Rokeya University, Rangpur

2nd Year | 1st Semester | Session 2023–2024
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Introduction & Motivation

Why autonomy matters in a compact educational platform

From manual control to independent motion.

Remote-controlled robots depend on an operator and communication range. This project develops a low-cost 2WD robot that senses its surroundings, chooses a safe path, and moves without supervision.

LEARNING FOCUS
Microcontroller interfacing • timing • control logic



Problem Statement

A robot must sense, decide, and act—without supervision

01 **SENSE** A single forward sensor has limited spatial awareness.

02 **DECIDE** The environment must be scanned to compare multiple paths.

03 **ACT SAFELY** Invalid echoes require a fail-safe response, not blind motion.

Objectives

Five concrete engineering outcomes

LOW-COST PLATFORM

Autonomous 2WD chassis

DISTANCE SENSING

HC-SR04 obstacle detection

LEFT / RIGHT SCAN

SG90 servo at 150° / 30°

PWM MOTOR DRIVE

L298N dual H-bridge

TURN MANEUVERS

90° turns and 180° recovery

Scope of the Work

Deliberately focused on robust indoor autonomy

INCLUDED

Hardware assembly
Wiring and firmware
Control algorithm
Fail-safe logic
Testing and documentation

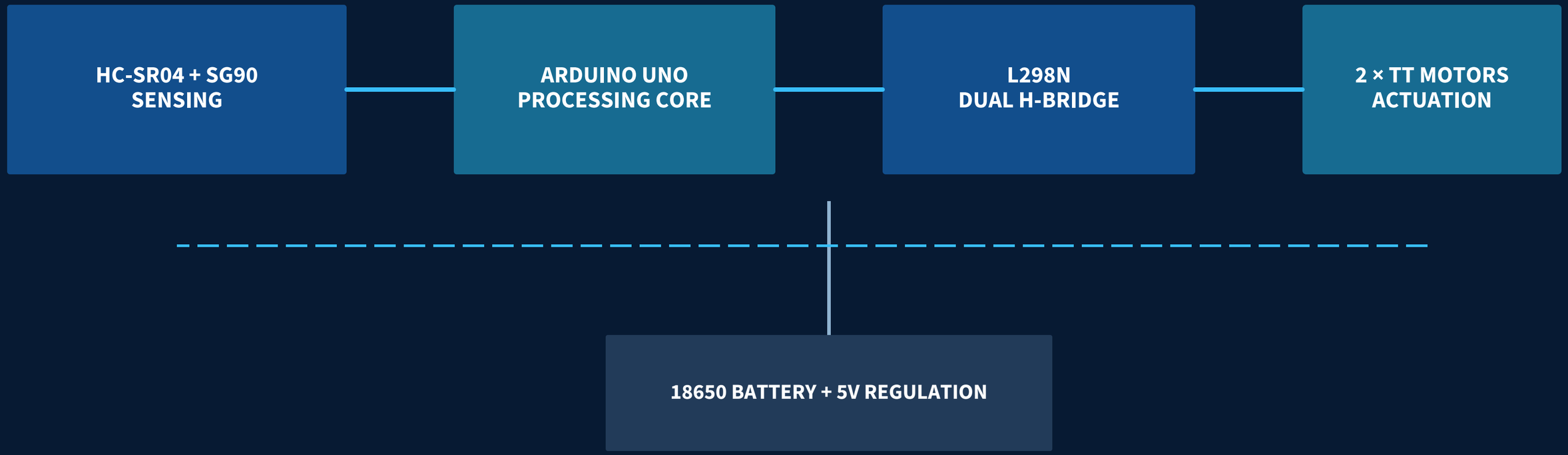
OUT OF SCOPE

SLAM
Computer vision
Advanced path planning
Wireless control

FOCUS / Low-cost autonomous obstacle avoidance in an indoor environment—with maximum educational value.

System Architecture

Sensing → processing → power stage → actuation



Hardware Components

Bill of materials and functional role

COMPONENT	QUANTITY	PURPOSE
Arduino Uno	1	Main controller
L298N Motor Driver	1	Motor control
TT Gear Motor	2	Robot locomotion
HC-SR04 Ultrasonic Sensor	1	Obstacle detection
SG90 Servo Motor	1	Sensor scanning
2WD Chassis	1	Robot frame
18650 Battery	1	Power source
Jumper Wires & Breadboard	Various	Connections
7805 Voltage Regulator	1	Stable 5V supply

Pin Mapping & Wiring

Deterministic I/O allocation for sensing and motor control

SIGNAL	ARDUINO PIN	NOTES
ENA	3	Motor A enable (PWM)
IN1	7	Motor A input 1
IN2	6	Motor A input 2
IN3	5	Motor B input 1
IN4	4	Motor B input 2
ENB	11	Motor B enable (PWM)
TRIG	9	Ultrasonic trigger
ECHO	8	Ultrasonic echo
SERVO	10	Servo control

WIRING NOTES

Remove ENA/ENB jumpers on L298N.

A common ground is mandatory across Arduino, driver, sensor, servo, and battery.

Overall Working Principle

A continuous sense–decide–act loop

01 SENSE

Trigger pulse → echo duration

Valid: 2–200 cm | Timeout: 12,000 µs

02 DECIDE

Distance = (duration × 0.0343) / 2

Obstacle when distance ≤ 30 cm

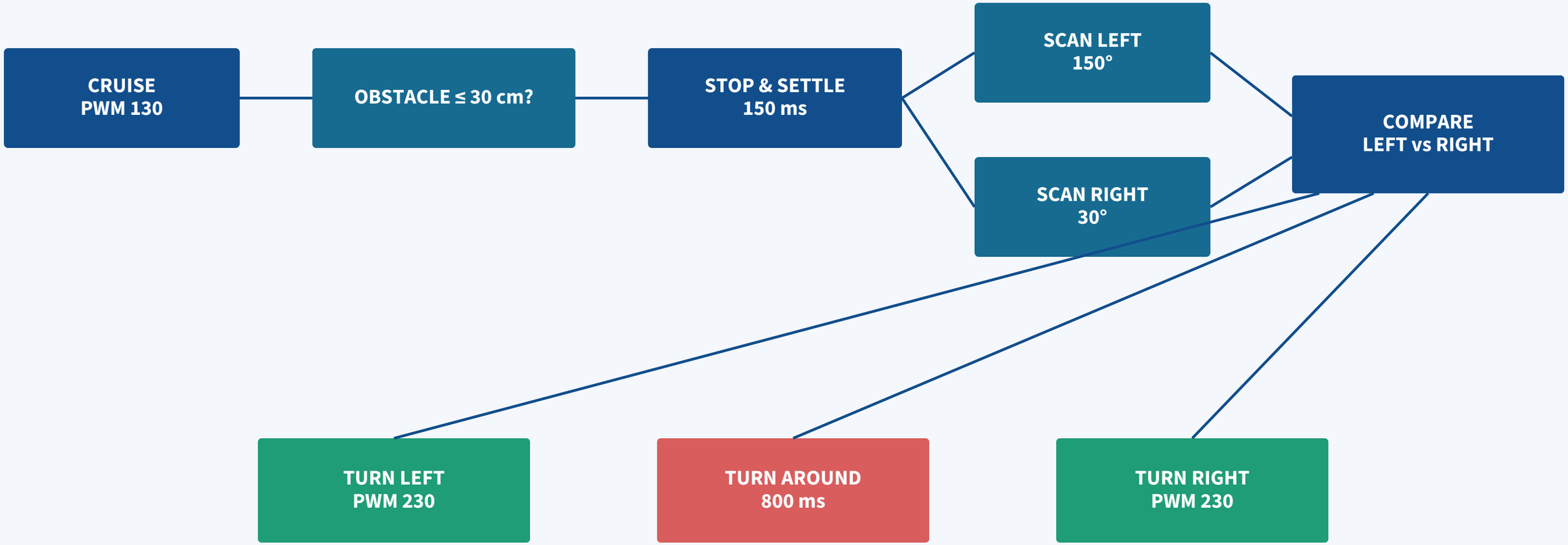
03 ACT

Forward • left • right • turnaround

PWM commands drive both motors

Control Logic / State Machine

Cruise at PWM 130; scan only when the forward path is blocked



Decision Logic & Tuning Parameters

All behavior is governed by explicit, testable constants

PARAMETER	VALUE	DESCRIPTION
NORMAL_SPEED	130	Cruising PWM
TURN_SPEED	230	Turn PWM
UTURN_SPEED	230	180° turn PWM
OBSTACLE_LIMIT	30 cm	Obstacle threshold
EQUAL_LIMIT	5 cm	Equal-side tolerance
SERVO_CENTER	90°	Center angle
SERVO_LEFT / RIGHT	150° / 30°	Scan angles
TURN_90_TIME	400 ms	90° duration
TURN_180_TIME	800 ms	180° duration

DECISION RULE
| Left − Right | ≤ 5 cm
→ execute 180° turnaround

Implementation / Software

Arduino IDE • Servo library • two-source-file structure

obstacle_avoiding_robot.ino
obstacle.cpp

Firmware separates sensing, decision-making, and actuation into small testable functions.

CONTROL CYCLE

```
setup()       initialize pins, servo, serial
loop()        main control cycle
getDistance() trigger, time, validate
scanSides()   acquire left / right distances
forward()     apply cruise PWM
stopCar()     zero motor outputs
turnLeft() / turnRight() / turnAround()
```

Results & Test Cases

All 10 test cases passed without false observations

10 / 10

TEST CASE	DESCRIPTION	EXPECTED BEHAVIOR	RESULT
Clear path	No obstacles	Forward motion	PASSED
Obstacle left	≤ 30 cm left	Turn right	PASSED
Obstacle right	≤ 30 cm right	Turn left	PASSED
Obstacle front	Both sides blocked	180° turnaround	PASSED
Boundary test	Exactly 30 cm	Correct turn	PASSED
Invalid < 2 cm	Sensor error	Fail-safe stop	PASSED
Echo timeout	No echo	Fail-safe stop	PASSED
Invalid side	Servo scan error	Fail-safe action	PASSED

System Output / Serial Monitor

9600 baud • live evidence for debugging and validation

```
> Front: 25 cm
> Front: INVALID
> Left: 40 cm
> Right: 35 cm
> Decision: Turn Left
```

VALID DISTANCE
Confirms echo timing and conversion.

INVALID READING
Triggers the fail-safe path.

SIDE SCAN + DECISION
Shows the evidence behind each maneuver.

A low-cost robot that can sense, decide, and recover.

ACHIEVED

Validated 2WD autonomy with ultrasonic sensing, servo scanning, PWM motor control, and fail-safe behavior.

NEXT

Multi-sensor array • encoders / IMU • PID speed control • wireless monitoring • grid-based planning

Thank you. Questions?